CIE AS PHYSICS (9702) • QUESTION PAPER

4.3 Density & Pressure

Syllabus Section: 4. Forces, Density & Pressure **Paper Type:** Structured Questions (Paper 2) **Total:** 11 Questions (125 Marks)

Question 1 Easy [8 marks]

Part (a) [1 mark]

State the word definition of density.

Part (b) [3 marks]

When calculating density both the equation and units are required.

(i) State the equation for density, defining all the terms used.

[1]

(ii) State the SI units for density and a non-SI unit which is commonly used.

[2]

Part (c) [2 marks]

For pressure caused by a solid, state

(i) the definition of pressure

[1]

(ii) the equation for pressure, defining the terms used.

[1]

Part (d) [2 marks]

Fig. 1.1 shows a measuring cylinder containing a liquid with depth *h* and cross-sectional area *A*.

Using the equations you stated in (b) and (c), show that pressure at a point in the liquid is $\Delta P = \rho g \Delta h$

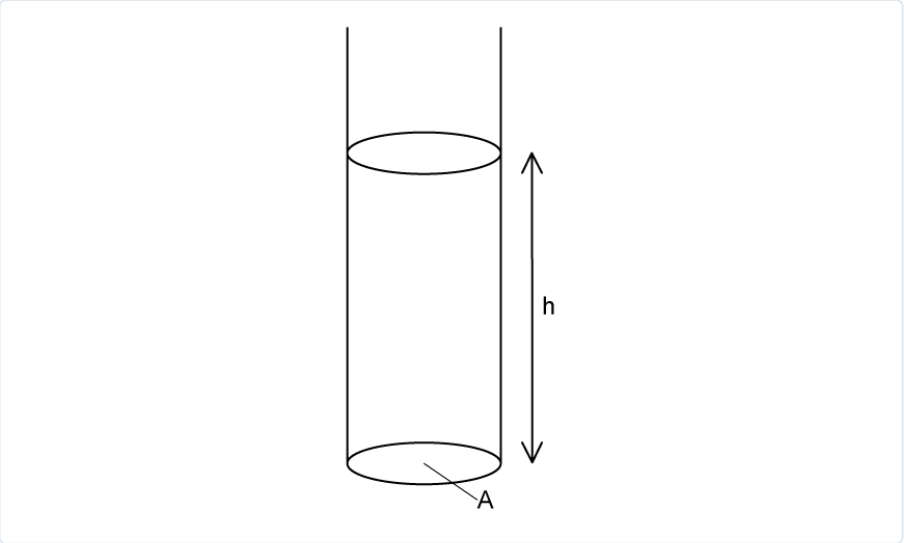


Fig. 1.1.

Question 2 Easy [13 marks]

Part (a) [5 marks]

The water in the measuring cylinder shown in Fig 1.1. has a depth *h* of 10 cm and a cross-sectional area *A* of 10 cm^2.

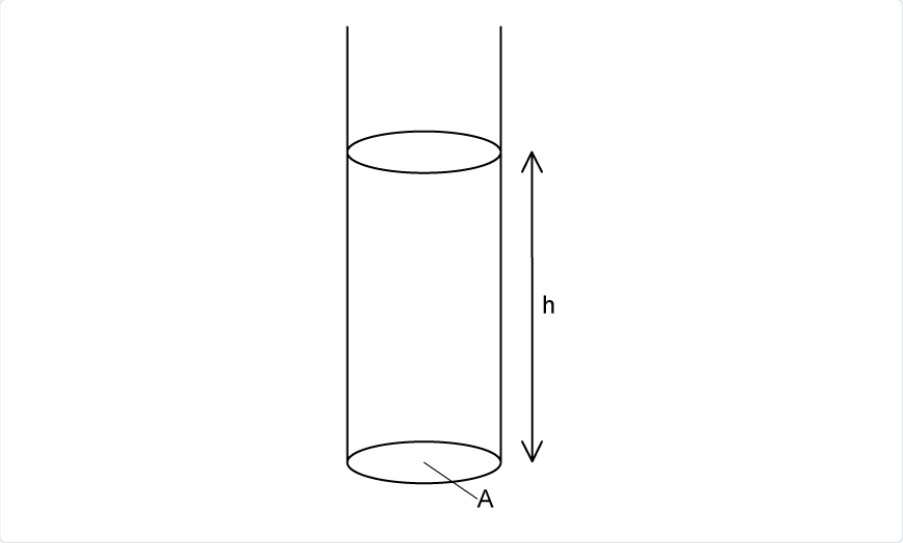


Fig 1.1.

(i) State the measurements of *h* and *A* in S.I. units.

[2]

(ii) Calculate the pressure at the base of the measuring cylinder due to the water. The density of water = 1000 kg m^{-3}.

[2]

(iii) Calculate the total pressure at the base of the measuring cylinder. Atmospheric pressure = 1.01×10^5 Pa.

[1]

Part (b) [2 marks]

The water pressure is measured at the base of two other measuring cylinders of different sizes.

One of the cylinders contains water with a depth five times larger than the cylinder in (a).

The other cylinder contains the same depth of water as the cylinder in (a), but the cross-sectional area of the base is nine times larger.

Without further calculation, state the effect on the water pressure at the base of the cylinder where

(i) the water is five times deeper

[1]

(ii) the surface area of the base is nine times larger.

[1]

Part (c) [3 marks]

A spherical ball is placed in the measuring cylinder so that it is submerged but floats in the water, as shown in Fig. 1.2.

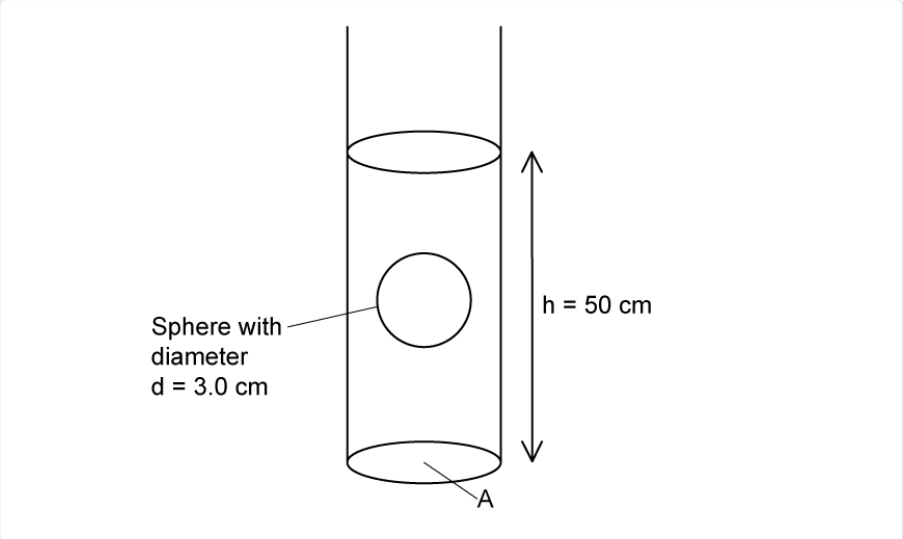


Fig. 1.2.

The ball has a diameter of 3.0 cm.

Calculate the volume of the ball in S.I. units.

Part (d) [3 marks]

Calculate the upthrust on the submerged ball in part **(c)**.

Question 3 Easy [12 marks]

Part (a) [3 marks]

Complete the definition of hydrostatic pressure.

Hydrostatic pressure is the pressure that is exerted by a _____ at _____ at a given point within the fluid, due to the force of _____.

Part (b) [2 marks]

A cylinder of length 1.8 m is submerged vertically in a tank of seawater so that the top of the cylinder is 0.2 m below the surface of the water, as shown in Fig. 1.1.

The density of the seawater is 1020 kg m^{-3}.

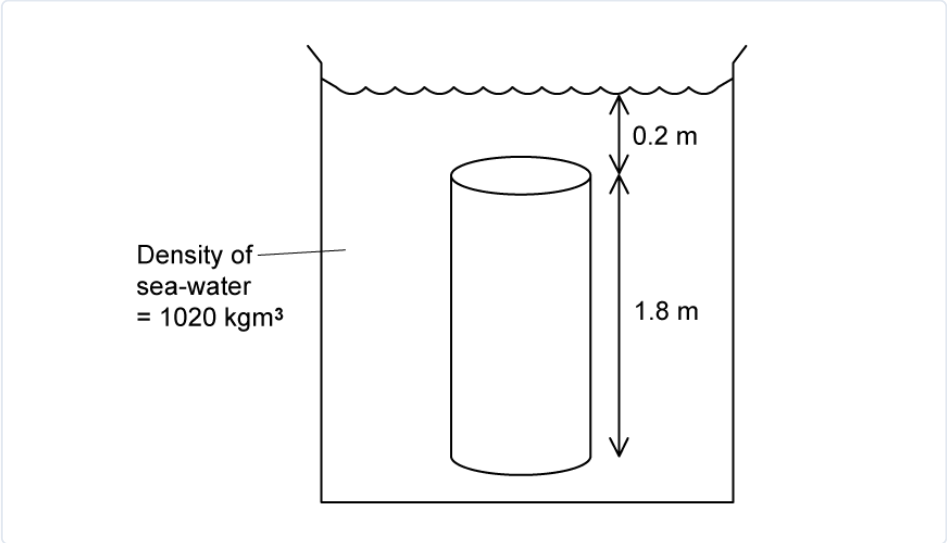


Fig. 1.1.

Calculate the difference in hydrostatic pressure between the top and bottom surfaces of the cylinder.

Part (c) **[4 marks]**

The cylinder has a diameter of 75 cm.

Calculate the upthrust on the cylinder.

Part (d) **[3 marks]**

The seawater in Fig. 1.1. is replaced with pure water which has a lower density than seawater.

Without further calculation

(i) State and explain the effect this would have on the upthrust on the cylinder

[2]

(ii) Hence, state whether the cylinder would sink, float or stay at the same position in the tank.

[1]

Question 4 Medium [18 marks]

Part (a) [6 marks]

A holiday company marketing 'unforgettable experiences' needs to know the maximum safe heights from which guests can jump into the sea from the end of a pier. Guests must be guaranteed they will not sink deep enough into the water to hit the bottom.

To investigate this, the designer tests how far into a measuring cylinder of water a wooden rod will sink when dropped from various heights.

The designer sets up their equipment as shown in Fig. 1.1.

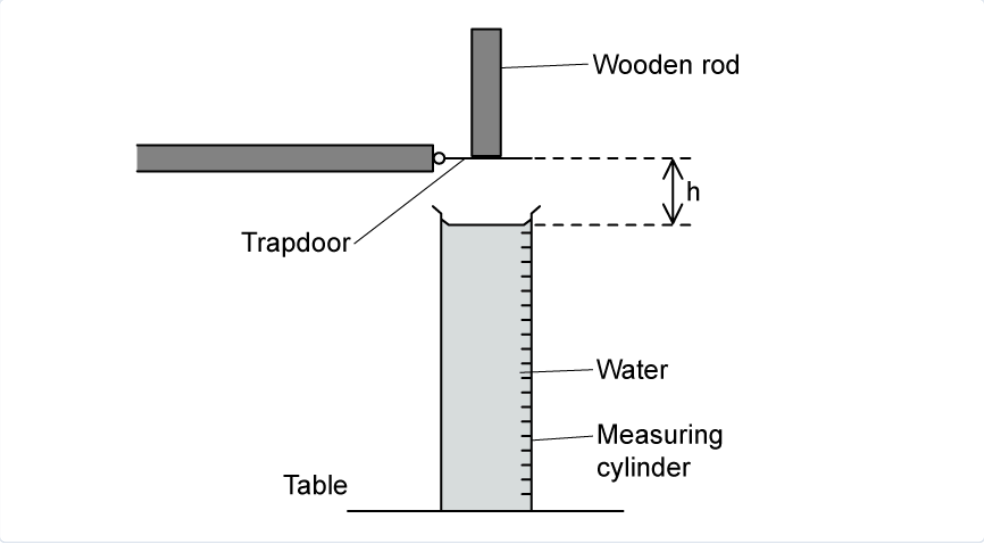


Fig. 1.1

The wooden rod has mass 1.30×10^{-2} kg, diameter 1.5×10^{-2} m and length 7.5×10^{-2} m.

(i) Calculate the density of the wood.

[2]

(ii) State and explain why the designer chose this type of wood to use in their investigation.

[2]

(iii) Suggest one change to the experimental set-up which the designer needs to make to model the situation they are investigating.

[2]

Part (b) [2 marks]

The wooden rod is held in a clamp so that it can be quickly released from rest.

Initially, it is clamped so that the base is at a height h = 0.20 m above the water surface.

Calculate the speed of the rod as its base reaches the water.

Ignore air resistance.

Part (c) [6 marks]

Fig. 1.2 shows the wooden rod fully submerged under the water surface.

The rod is moving vertically downwards and has not yet come to rest.

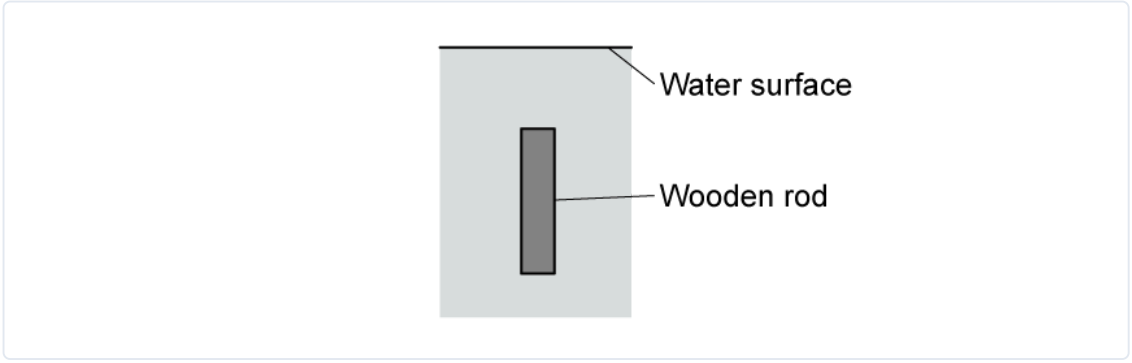


Fig. 1.2

(i) Sketch on Fig. 1.2 the forces acting on the wooden rod.

[3]

(ii) Describe and explain how the resultant force on the wooden cylinder varies from the moment the cylinder is fully submerged until it reaches its deepest point.

[3]

Part (d) **[4 marks]**

The investigation concludes that an exciting but safe experience can be had by jumping into the water which is at least six metres deep. The designer suggests a minimum age of 15 years for guests taking part in the experience.

The density of the seawater is 1030 kg m^{-3}.

Assuming that guests will sink to no more than 4 metres, jumping feet first, estimate the range of pressure they will feel in their ears.

Question 5 Medium [9 marks]

Part (a) [2 marks]

Party shops sell helium-filled balloons for various events. One such balloon is shown in Fig. 1.1.

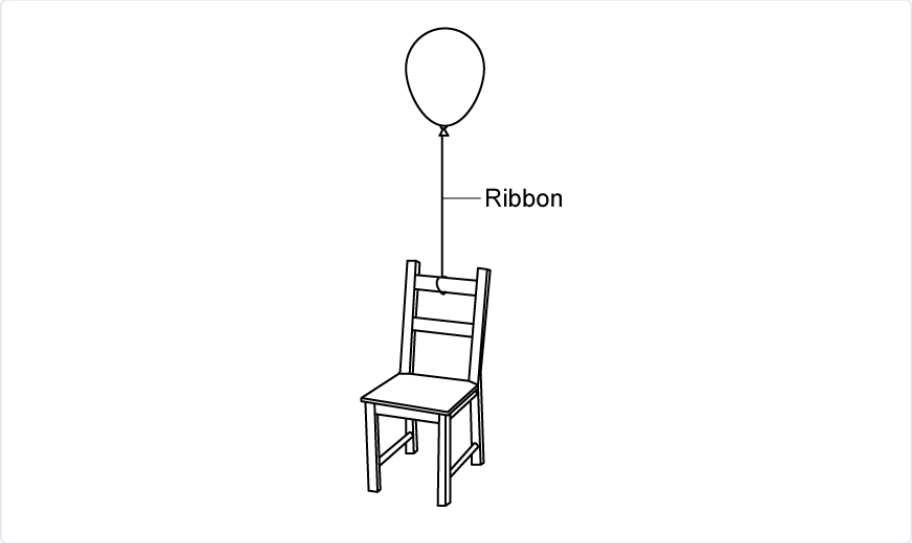


Fig. 1.1

The ribbon is tied to a chair to stop the balloon floating away.

By modelling the balloon as a point mass, sketch a free body force diagram for the balloon and ribbon.

Part (b) [3 marks]

At its widest point, the balloon has a diameter 40 cm. The density of the surrounding air is 1.30 kg m^{-3}.

Calculate the upthrust on the balloon.

Part (c) **[4 marks]**

Balloons filled with air do not float.

Explain why a balloon filled with helium floats.

Question 6 Medium [12 marks]

Part (a) [4 marks]

The Galileo thermometer was invented in the 17th century and it is still popular today. The thermometer consists of a column of liquid containing roughly spherical, weighted, glass bulbs, as shown in Fig. 1.1.

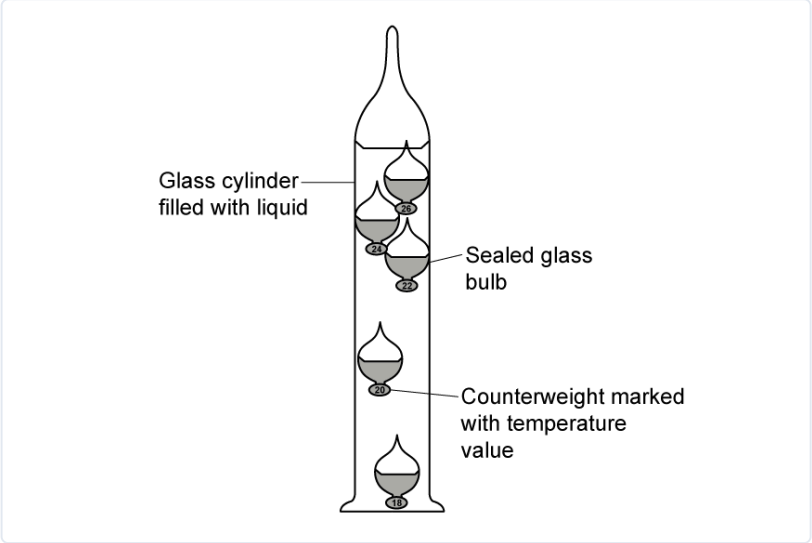


Fig. 1.1

The spheres move up or down in the column as the temperature changes.

At a certain temperature, one of the spheres begins to fall through the liquid at an increasing speed.

(i) State the property of the liquid that causes the spheres to move up or down as the temperature changes.

[1]

(ii) The sphere has a radius of 1.2 cm and once settled, it displaces a total mass of 7.3 g of the liquid.

Calculate the value of the property of the liquid named in **(a)(i)**.

[3]

Part (b) [4 marks]

Sketch arrows on Fig. 1.2 to show the relative sizes of the forces acting on the sphere as it falls.

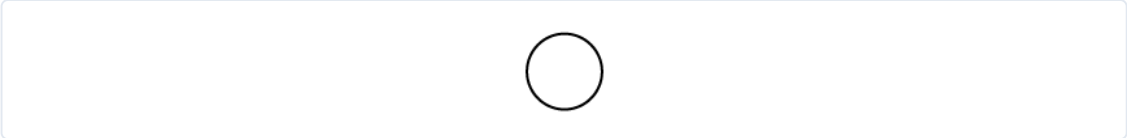


Fig. 1.2

Part (c) [2 marks]

Show that the upthrust on the sphere is about 0.07 N.

Part (d) [2 marks]

The spheres in the column are identical but have different masses due to the counterweight attached to them.

State and explain where you would expect to find a sphere of weight 0.069 N.

Question 7 Medium [9 marks]

Part (a) [2 marks]

A submarine below the surface of the sea is stationary in a region where the density of the seawater is 1.03×10^3 kg m^{-3}.

The submarine has a volume of 5.83×10^3 m^3.

Calculate the upthrust exerted on the submarine by the seawater.

Part (b) [2 marks]

Explain why the mass of the submarine must be 6.0×10^6 kg.

Part (c) [3 marks]

The submarine moves into a region of the sea where the water is less salty, and the density of the water reduces to 1.01×10^3 kg m^{-3} .

Explain what would happen to the submarine as it moves into this region of lower density seawater.

Part (d) [2 marks]

The submarine alters its weight by pumping water in or out of its internal tanks.

> Determine the mass of water that the submarine should pump, in or out of its tanks, to maintain its depth below the surface of the sea.

Question 8 Medium [9 marks]

Part (a) [2 marks]

A hollow brass cylinder with closed ends is floating on the surface of the water.

The cylinder has a length of 4.0 cm and an external diameter of 2.1 cm as shown in Fig. 1.1.

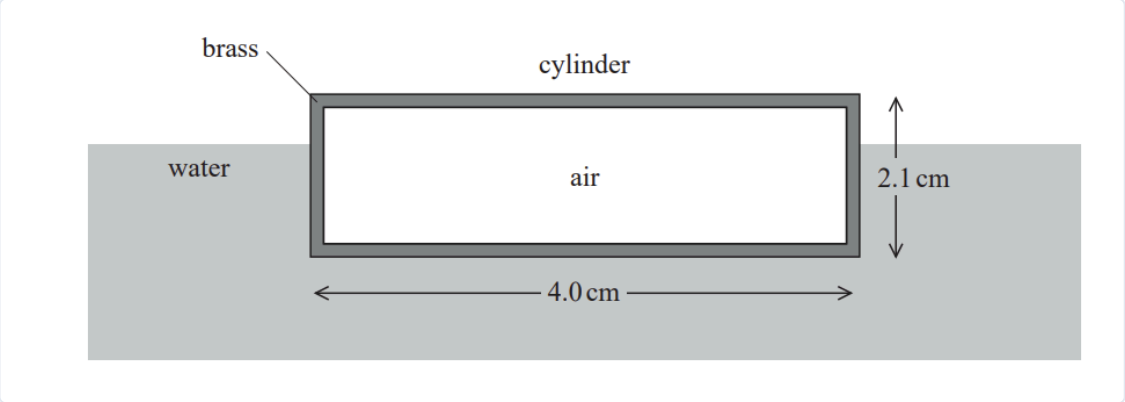


Fig. 1.1

63% of the volume of the cylinder is submerged. The cylinder contains negligible weight of air.

Explain why the brass cylinder floats.

Part (b) [3 marks]

The density of water is 1.0×10^{3} kg m^{-3}.

Show that the mass of the cylinder is about 9×10^{-3} kg.

Part (c) [4 marks]

Deduce whether an identical hollow cylinder made of gold would also float.

Assume that the volume of gold is the same as the volume of brass.

- Density of gold = 19.3×10^3 kg m^{-3}
- Density of brass = 8.7×10^3 kg m^{-3}

Question 9 Hard [10 marks]

Part (a) [6 marks]

The Cartesian Diver is a popular science project which can be used to demonstrate physical concepts.

A 'diver' is made by bending a loop of drinking straw into a U-shape and then weighting it with modelling clay.

A large plastic bottle is filled with water, leaving a small volume of air at the top.

The diver is carefully added, so that some water enters it but the straw remains floating upright due to trapped air.

Finally the lid is tightly screwed back onto the bottle, making a sealed container.

The completed Cartesian Diver and the construction of the diver are shown in Fig. 1.1.

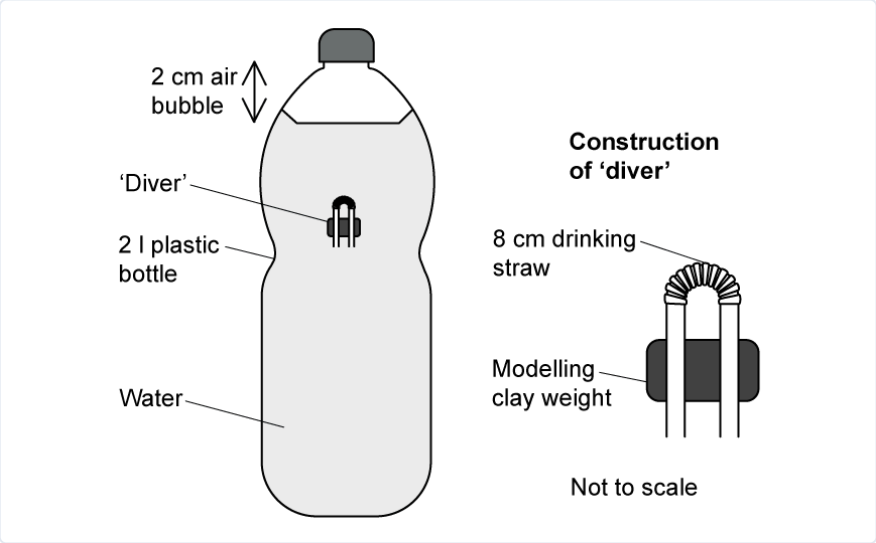


Fig. 1.1.

When the sides of the bottle are squeezed and released in response the 'diver' moves up and down rapidly as illustrated by Fig. 1.2.

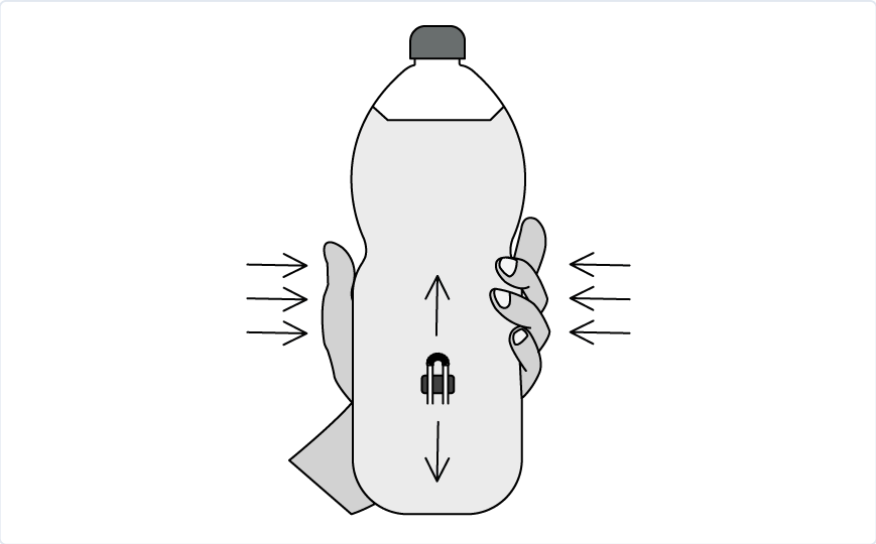


Fig. 1.2.

Discuss the physics of the Cartesian Diver when the sides of the bottle are squeezed.

(i) Explain the observed motion of the 'diver'.

[5]

(ii) Predict which direction the diver moves in when the sides of the bottle are squeezed.

[1]

Part (b) **[4 marks]**

When the sides of the bottle are released the diver reverses direction.

Explain why the diver returns to its original position in the bottle.

Question 10 Hard [13 marks]

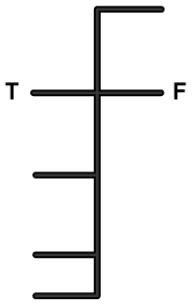
Part (a) [4 marks]

To prevent the overloading and sinking of ships markings, known as load lines, are required on ships hulls since the Middle Ages.

As the ship is loaded it sinks lower in the water. Once the load line meets the waterline, the ship is deemed to be fully loaded.

In 1876 Samuel Plimsoll refined the load line to reflect the needs of increasingly international trading routes. Today the 'Plimsoll line' has markings to show safe loading levels in fresh, saltwater, and warm and cold temperatures.

The Plimsoll line markings on ships today are shown in Fig. 1.1. The expected labels, their meanings and indications of temperature are written below.



F - Fresh water
T - Tropical seawater (warm)
S - Summer temperate seawater (cool)
W - Winter temperate seawater (cold)
TF - Tropical fresh water (very warm)
WNA - Winter North Atlantic (extreme cold)

Fig. 1.1.

Explain why the ships sit lower in the water when they are fully loaded.

Part (b) [7 marks]

With reference to upthrust

(i) Explain why different load lines for types of water and times of year are useful.

[5]

(ii) Hence complete the diagram in Fig. 1.1. by adding the missing labels S, TF, W and WNA.

[2]

Part (c) [2 marks]

In a simple classroom demonstration of the principle behind the Plimsoll Line, a teacher places an egg in a beaker of distilled water at close to 0°C.

The egg sinks to the bottom of the beaker as shown in Fig. 1.2.

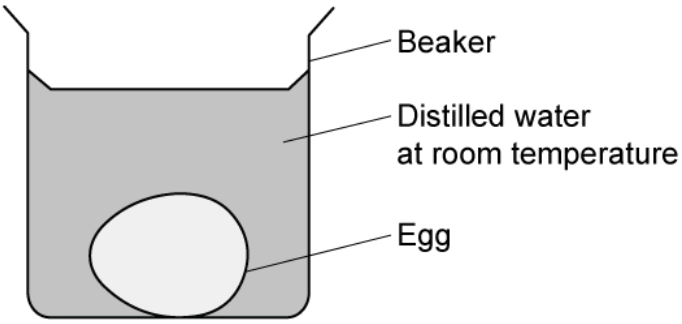


Fig. 1.2.

Suggest and comment on the likely next steps in this demonstration.

Question 11 Hard [12 marks]

Part (a) [4 marks]

Hot air balloons consist of an envelope with an open mouth through which air can be heated using a burner. The air inside becomes much hotter than its surroundings. Lifted by this method, the balloon can carry a large amount of weight across very long distances.

A very large capacity balloon, as shown in Fig. 1.1. can carry 25-30 people.

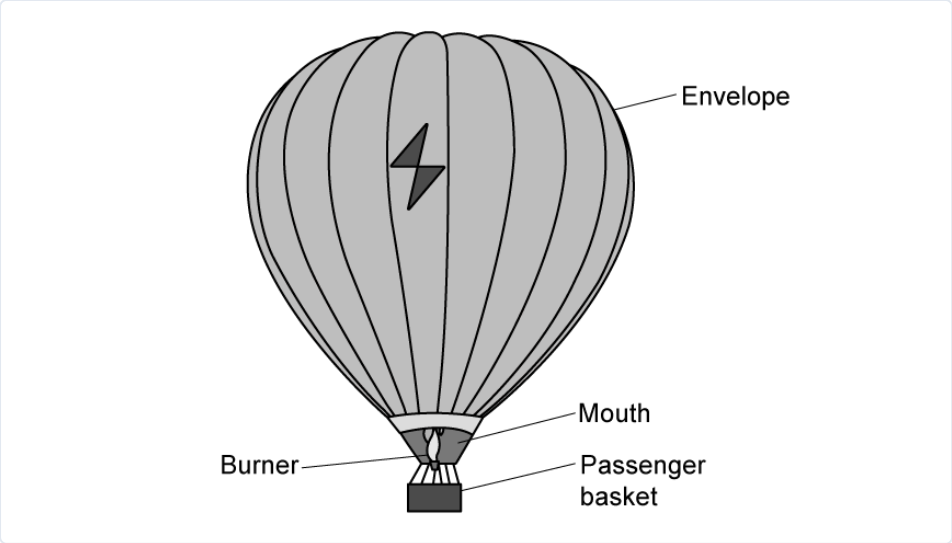


Fig. 1.1.

The envelope has a diameter at its widest point of of 32 m when it is filled with air. The total mass of the balloon, air, equipment and passengers is 20 900 kg. When the air temperature is 15 °C the density of the air outside the balloon is 1.225 kg m^{-3}.

When the air inside the envelope is hot enough, the balloon is released.

(i) Calculate the resultant upwards force on the balloon.

[3]

(ii) Calculate the initial upwards acceleration of the balloon.

[1]

Part (b) **[4 marks]**

The balloon is taken to the Middle East, where the temperature is 45 °C.

Discuss how this would affect the ability of the balloon to fly and any changes the pilot could make to fly in these temperatures.

Part (c) **[4 marks]**

The air density at 45 °C is 4 % less than the air density at 15 °C.

For the hotter conditions

(i) Calculate the mass that must be removed in order to achieve the same initial acceleration as in part (a)(ii)

[3]

(ii) Suggest how this might be achieved

[1]